

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
1	(a)	(i)	0.4 [Ω] (1) Contact resistance / resistance of connecting wires and or crocodile clips (1) Don't accept internal resistance / natural resistance		1	1	2	1	2
		(ii)	Any 3 × (1) from: • Straight line • Positive gradient / as length increases, resistance increases • Points close to line of best fit / little scatter in the points • Positive intercept so doesn't agree / if 0.4 subtracted line would go through origin so agreement • Graph neither confirms nor denies dependency on ρ or A but is consistent with constancy of $\frac{\rho}{A}$ Award no marks for directly proportional			3	3		3
		(iii)	[Digital] callipers / micrometer (1) Accept screw gauge 0.01 mm unit mark (1)	2			2		2
		(iv)	Gradient determined to be 11.6 ± 0.1 or 0.116 ± 0.001 (1) Gradient = $\frac{\rho}{A}$ (can be implied) (1) Area = $\pi(0.115 \times 10^{-3})^2 = 4.2 \times 10^{-8} \text{ [m}^2\text{]} (1)$ $\rho = 4.9 \times 10^{-7} \Omega \text{ m}$ ecf on gradient and area (1) unit mark -1 mark for arithmetical slips of power of 10 / factor of 2 slip Alternative for first and second marking points: Use of point from line to give R value with 0.4 Ω subtracted and corresponding l value (1) Substituted into $\rho = \frac{RA}{l}$ (1)			4	4	4	4